

UCT41042 – Practical for Introduction to Smart System
Department of Information & Communication Technology
Faculty of Technology
South Eastern University of Sri Lanka
Lab Sheet No: 06

Date: 18.05.2026

Title: Programming ESP32 to Interface Input & Output Devices using Wokwi

Aims: To learn how to program the ESP32 to correctly read data from input devices and control output devices using the Wokwi online simulator, while understanding the purpose and behavior of each device.

Tasks:

Input Devices

1. Push Button (Tactile Switch)

Write an ESP32 program to:

- Detect button press
- Print “Button Pressed” and “Button Released” on Serial Monitor.

2. Slide Switch / Toggle Switch

Write a program to:

- Read the switch state
- Display ON or OFF in Serial Monitor

3. Potentiometer (Rotary)

Write a program to:

- Read the analog value from GPIO 34
- Print values (0–4095) on Serial Monitor

4. Rotary Encoder (KY-040)

Write a program to:

- Detect clockwise and anticlockwise rotation
- Print direction and step count

5. Analog Joystick

Write a program to:

- Read X-axis, Y-axis, and button press
- Print values in Serial Monitor

6. Keypad

Write a program to:

- Detect key presses
- Display pressed key on Serial Monitor

7. IR Receiver

Write a program to:

- Receive IR remote signals
- Print received HEX code

Output Devices

1. IR Remote

Observe how:

- Button press sends IR signal
- Signal is decoded by ESP32

2. LED

Write a program to:

- Blink LED with 1-second delay

3. RGB LED

Write a program to:

- Display Red, Green, Blue, and White colors

4. Buzzer

Write a program to:

- Generate beep sound using GPIO

5. Servo Motor

Write a program to:

- Rotate servo from $0^\circ \rightarrow 90^\circ \rightarrow 180^\circ$

6. Bipolar Stepper Motor + A4988 Driver

Write a program to:

- Rotate stepper motor clockwise and anticlockwise

7. Relay Module

Write a program to:

- Turn relay ON and OFF using ESP32

8. LCD 16x2 (Parallel)

Write a program to:

- Display your name and roll number

9. LCD 16x2 (I2C)

Write a program to:

- Display sensor values on LCD

10. MAX7219 LED Dot Matrix

Write a program to:

- Display scrolling text

11. Seven Segment Display

Write a program to:

- Display digits 0–9 sequentially